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Intelligent ESG portfolio optimization: A multi-objective AI-driven framework for sustainable investments in the Indian stock market

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ABSTRACT

Environmental, Social, and Governance (ESG) portfolio management impacts the investor in Sustainable and Socially Responsible Investments (SRIs). However, existing frameworks often fail to integrate artificial intelligence (AI) to optimize ESG portfolios, particularly in the Indian stock market. They are lacking in terms of achieving higher efficiency in predictive returns and less efficient in integrating ESG scores with portfolio optimization and subsequently rebalancing the optimum portfolio returns. Hence, this study proposes a three-stage AI-driven framework to effectively manage ESG portfolios while maximizing predictive accuracy and investment efficiency. It demonstrates on 30 randomly selected ESG-ranked Indian stocks from diverse sets, and simulates a retail portfolio scenario by leveraging advanced machine learning and optimization techniques. In first stage, a Multivariate Bidirectional Long Short-Term Memory (MBi-LSTM) network is utilized to enhance return prediction accuracy, capturing the market's nonlinear dynamics. In the second stage, portfolio allocation is optimized by balancing risk, return, and ESG scores using the Non-dominated Sorting Genetic (NSG) algorithm. In the final stage, dynamic portfolio rebalancing is performed using the Actor-Critic Reinforcement Learning (RL) algorithm by integrating ESG scores. For predictive modeling, MBi-LSTM showed the lowest errors in all the parameters. The NSG-enabled portfolio allocation method achieved 68.58 % higher similarity to actual returns than the Mean-Variance Markowitz (MVM) model while maximizing the ESG-Sortino Ratio. In portfolio rebalancing, it outperformed traditional agent-based methods, with returns higher by 120 %, 235 %, and 101 % in average, maximum, and minimum returns, respectively. The effectiveness of the demonstrated Multi-Objective AI-driven framework is showing in every stage validates against the conventional evaluation metrics, showcasing its superior performance and practical viability for sustainable and SRI investment in the Indian stock market. Hence, this framework can help the investor to manage their ESG portfolio for a better SRI.

1. Introduction

Sustainable and Socially Responsible Investments (SRIs) have gained popularity among investors in India [1]. Sustainable investments incorporate Environmental, Social, and Governance (ESG) factors, such as carbon emissions, human rights commitments, and corporate governance practices. ESG integration helps investors mitigate risks, achieve competitive returns, and align with long-term financial and ethical goals. Strong ESG practices enhance stability, reduce regulatory risks, and attract growing market demand. SRIs, a subset of sustainable investments, focus on social and ethical considerations, including labor standards, human rights, and community development [2]. By incorporating ESG principles, investors can build resilient portfolios while supporting responsible and sustainable businesses.

The popularity of SRIs in India is reflected in the increasing number of SRI funds lunched by Indian asset managers. According to a [3], ESG is being increasingly integrated into investment selection and due diligence processes. The assets under management (AUM) of ESG funds in India surged by around 150 %, from $275 million in 2020 to $650 million in 2021, surpassing $1 billion by 2023. Additionally, the Nifty ESG Index has outperformed the benchmark Nifty in recent years, demonstrating the growing acceptance of ESG-focused funds. This trend is anticipated to persist, with more investors likely to choose SRI funds as become more aware of the financial and social impact of their investments. This shift could lead to a more socially conscious and sustainable economy in India [4,5].

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With the growing emphasis on sustainability and responsible corporate practices, India has also made significant progress in ESG regulations [6]. In 2021, SEBI mandated the top 1000 listed companies to disclose ESG-related information, promoting transparency and accountability [7]. Standardized guidelines for sustainability reporting were introduced, enabling better ESG assessment through metrics such as carbon emissions, board diversity, and ethical practices. Rating agencies like CRISIL, ICRA, and CARE have developed ESG frameworks to help investors evaluate corporate sustainability [8]. These advancements have supported the rise of SRI portfolios in India by allowing the systematic integration of ESG factors into investment strategies. While challenges remain in scoring consistency, the availability of standardized ESG data has enhanced investor decision-making and reaffirmed India's commitment to ethical investing [9]. Moreover, it becomes a pivot point for the investors to optimize their ESG portfolio for better SRIs with balanced risk and returns. Hence a framework is needed for the investor which can achieving higher efficiency in predictive returns and efficiently integrate the ESG scores with portfolio optimization and subsequently rebalancing the optimum portfolio returns.

Many studies emphasize the critical role of ESG criteria in making investments decision, linking successful investments to sustainability through modern machine learning (ML) methods. For example, [10] was using Natural Language Processing (KoBERT) in Korea found abnormal trading volumes on days with ESG controversies, indicating that such controversies influence investors behavior. They also noted that the impact varies by ESG pillar, with institutional investors often liquidating securities in response to ESG controversies, suggesting that positive ESG practices benefits both the firm and investors [11]. found a negative relationship between ESG performance and stock price fragility, meaning better ESG performance reduces stock price volatility and investors sensitivity [12] . showed that high ESG scores lead to higher equity returns and lower financing cost. They also noted that ESG risks are incorporated into bond pricing, with ESG factors significantly impacting bond yields [13]. concluded that ESG combined score correlates with firm value. These studies collectively demonstrate that good ESG scores benefit firms, investors, and society, reinforcing the importance of ESG in investment strategies.

Numerous studies have applied ML techniques to investment optimization, ranging from traditional statistical models to advanced deep learning architectures. Prominent ML algorithms explored for portfolio optimization include Artificial Neural Networks (ANN), Support Vector Regression (SVR), and Long Short-Term Memory (LSTM) networks. ANN, inspired by the human brain, excel at capturing complex nonlinear relationships in financial forecasting but often lack stability for long-term strategies, as shown by [14]. SVR has also shown suboptimal performance, sometimes yielding negative returns, making it unsuitable for medium- and long-term forecasting [15,16]. In contrast, LSTM networks, a type of recurrent neural network (RNN), effectively capture temporal dependencies in financial markets, demonstrating superior accuracy and robustness over extended periods. This makes LSTM a powerful tool for reliable investment forecasts. Other ML algorithms, such as Random Forests and Gradient Boosting Machines (GBM), offer unique advantages in predictive accuracy and computational efficiency but struggle to adapt dynamically to changing market conditions. While these algorithms handle nonlinear relationships well, they typically result in static portfolios. Overall, LSTM networks have proven the most promising for financial modeling and investment optimization due to their ability to retain and selectively update information, capturing intricate patterns and trends to enhance predictive capabilities beyond traditional ML algorithms [17].

Appling LSTM for prediction and portfolio optimization faces challenges. First, SRI strategies often balance financial returns and ESG targets, requiring models to adapt to market changes, evolving ESG priorities, and new ESG risks and opportunities. Standard LSTM models may struggle to keep pace with these variables. Additionally, univariate LSTM models may fail to capture relationships across different ESG metrics, financial data, and market indicators, limiting their ability to make well-informed investment decisions. To address these issues, we propose a Multivariate Bidirectional LSTM (MBi-LSTM) model. This model handles multivariate data, capturing interactions across multiple variables and enhancing feature learning by recognizing patterns across various dimensions. Its bidirectional nature strengthens the understanding of temporal dependencies by considering past and future contexts, enabling robust feature extraction and capturing subtle trends in the data. This MBi-LSTM addresses dynamic adaptability and interdependency challenges. A similar model, discussed in a study by [18], uses a Deep Reinforcement Investment Portfolio (DRIP) Model based on MBi-LSTM, showing dynamic adaptability in retraining prediction models and rebalancing portfolios effectively. Our research explores how the MBi-LSTM performs in the Indian context, comparing its predictive capacities to other models, including Gated Recurrent Units (GRU), ARIMAX, and Ensemble Models combining Generalized Autoregressive Conditional Heteroskedasticity (GARCH) and Autoregressive Integrated Moving Average (ARIMA) [15,17].

The study aims for a framework that can foster superior portfolio performance and practical viability for multi-objective portfolio optimization in the Indian market, potentially paving the way for the development of self-monitoring intelligent portfolios capable of automatic investment decision-making. The first objectives of this research to predict the future stock prices, returns and risks of 30 (randomly selected stocks of the *CRISIL 2022 ESG Rating report*) individual stocks, using machine learning algorithms along with an exploration for the best suited prediction mechanism. Second, to construct a portfolio that maximizes the custom fitness function comprising the return, risk and ESG constraints and rebalances itself dynamically based on changing market environment. Finally, we evaluate the proposed model on a potential portfolio of selected 30 Indian stocks over a period of one year (2022) to test the efficiency and output of the developed mechanism compared to conventional and unautomated method.

The study found that for the predictive modeling, MBi-LSTM showed the lowest errors in all the parameters such as mean absolute error (MAE), root mean squared error (RMSE), mean absolute percentage error (MAPE), and mean squared percentage error (MSPE). For the portfolio allocation, NSG-enabled method achieved 68.58 % higher similarity to actual returns than the Mean-Variance Markowitz (MVM) model while maximizing the ESG-Sortino Ratio. Finally, for in portfolio rebalancing, it outperformed traditional agent-based methods, with returns higher by 120 %, 235 %, and 101 % in average, maximum, and minimum returns, respectively. The effectiveness of the Multi-Objective AI-driven framework is validated at every stage against conventional evaluation metrics, demonstrating its superior performance and practical viability for sustainable and SRI investment in the Indian stock market. Hence, this framework can help investors efficiently manage their ESG portfolios for a better SRI.

The rest of this paper is organized as follows: Section 2 describes the methodology and framework design used in our analysis. Section 3 describe experimental test with different parameters. Section 4 contains prediction results and discussions behind the results. Section 5 explains the conclusions of the findings and finally Section 6 directs the future scope of our research.

2. Framework design

2.1. Predictive modeling

The first phase of the framework focuses on forecasting returns and risk through advanced predictive modeling techniques. Multiple models are employed and compared to identify the most accurate and robust forecasting approach. These models include GRU, Ensemble Models, Seasonal Autoregressive Integrated Moving Average with Exogenous Regressors (SARIMAX), and Multivariate Bidirectional Long Short-Term Memory (MBi-LSTM) networks [19], and others. By leveraging a diverse

set of predictive models, the research aims to capture the complex and nonlinear relationships inherent in the Indian stock market data, thereby enhancing the accuracy of return and risk forecasts.

2.2. Portfolio weights allocation

In the second phase, portfolio weights allocation is performed on a diversified 30-stock portfolio using the Non-dominated Sorting Genetic Algorithm (NSGA) [20]. The NSG algorithm is chosen for its ability to optimize portfolios based on a multi-objective fitness function, which maximizes the ESG-Sortino Ratio. The ESG-Sortino Ratio is a modified version of the traditional Sortino Ratio, incorporating ESG considerations to prioritize downside risk management while maximizing returns. By integrating ESG factors into the portfolio optimization process, the research seeks to construct portfolios that not only deliver superior financial performance but also align with socially responsible investment principles.

2.3. Reinforcement learning deployment

The final phase of the framework involves the deployment of RL techniques for periodic portfolio rebalancing [18]. Specifically, the Actor-Critic method adaptively adjust portfolio compositions in response to changing market dynamics and environmental variables in the Indian capital markets. By continuously learning from feedback loops and optimizing actions based on predefined objectives, the RL agent keep the portfolio resilient and responsive to evolving market conditions. This dynamic rebalancing mechanism enhances the portfolio's robustness and adaptability, mitigating potential risks and maximizing long-term returns.

2.4. Integration of phases

The three-phase framework operates sequentially, with each phase building upon the insights and outputs from the preceding stage. Predictive modeling results provide accurate forecasts of returns and risk metrics to inform the portfolio allocation process. Likewise, portfolio weights determined through the NSG algorithm serve as input for the RL agent and guide during portfolio rebalancing. This integrated approach ensures the portfolio optimization strategy remains holistic, adaptive, and aligned with the objectives of maximizing risk-adjusted returns while incorporating ESG considerations.

3. Methodology

3.1. Predictive modelling

3.1.1. Comparison between prediction models

In this research, we conducted the comprehensive assessment of various ML models to forecast stock prices for the year 2022 based on a decade-long dataset from 2011 to 2021. The ML models include U-LSTM, SVR, ARIMA, GRU, SARIMA, and an Ensemble Model. We evaluated each model based on key performance metrics such as MAE, RMSE, MAPE, and MSPE. Each model employed in our research has its own strengths and limitations in the realm of stock price prediction.

U-LSTM, renowned for capturing long-term dependencies in historical prices, struggles to incorporate external factors such as economic data or company news. SVR excels with non-linear relationships and limited data but faces challenges in model interpretation and hyperparameter optimization, due to the stock market's complexity. ARIMA is effective for stationary time series data but may falter due to its assumption of linearity and normality in stock price movements. GRU is akin to LSTM, learns long-term dependencies with fewer parameters but might struggle with capturing intricate relationships, particularly in volatile markets like India's. SARIMA effectively incorporates seasonal components; however, it is constrained by ARIMA's linearity and normality assumptions. The Ensemble Model captures both volatility and trend but poses challenges in computational complexity and interpretability.

MBi-LSTM stands out due to its ability to consider multiple factors alongside historical prices, crucial in India's information-rich market (see Table 1). It captures long-term dependencies and external factors like economic data and company news. Moreover, the bidirectional nature of the MBiLSTM processes data in both directions, enabling a comprehensive assessment of potential upcoming fluctuations compared to other models [21].

Despite the Ensemble Model ranking as the second-best performer, combining GARCH (to capture volatility) and ARIMA (to capture trends), it encounters challenges with large prediction horizon, requiring extensive training datasets. This becomes problematic for many stocks, especially with the recent surge in initial public offerings (IPOs) over the last decade, limiting the availability of extensive historical data. While the ensemble model offers robust predictive capabilities, its dependence on substantial training data poses practical constraints, making it less suitable for the current research's objectives. Therefore, the MBi-LSTM emerges as the preferred choice for stock price prediction in this research, offering a blend of accuracy, adaptability, and suitability for dynamic market conditions [22,23].

3.1.2. Implementation

This research implements an MBi-LSTM neural network to forecast 'Adj Close' stock prices using features including 'Open', 'High', 'Low', 'Close', 'Volume', and 'Adj Close. The model processes input sequence bidirectionally, capturing dependencies from both past and future data. The architecture includes multiple Bi-LSTM layers with Leaky ReLU activation, dropout layers for regularization, and a final dense layer for regression (see Fig. A1 in Appendix).

Key steps include data preprocessing, normalizing input features with Min-Max scaling and creates time series sequences using 'TimeseriesGenerator' from the Keras library. The model targets 'Adj Close' for prediction, training with Adam optimizer (learning rate 0.001 and clipnorm 0.01) and using MSE as the loss function. Early stopping prevent overfitting. Predicted values are combined with original test data, and performance metrics such as MAE, RMSE, MAPE, and MSPE are calculated using scikit-learn functions.

The implementation of a MBi-LSTM in stock prediction marks a significant shift, capturing bidirectional dependency inherent in financial data. Traditional models struggle with such complexities, but Bi-LSTM processes both historical trends and anticipatory signals effectively. This approach is crucial in financial markets, where decisions rely on the interplay of past and future information. The model's ability to process bidirectional data enhances its accuracy and generalization, making it a potent tool for stock price forecasting. Evaluation of the model's performance is done by comparing predicted stock price with actual value metrics such as MSE or MAE. A rigorous comparative analysis against existing models underscores the Bi-LSTM's effectiveness in stock prediction.

3.2. Portfolio allocation: Non-Dominated Sorting Genetic Algorithm (NSGA)

3.2.1. NSGA implementation for allocation

Our NSGA implementation utilizes Python and the Pandas library for data manipulation and NumPy for numerical computations. Following is the break-down of the code and its architecture:

3.2.1.1. Data loading and preprocessing. Initially, historical, and predicted stock price data are loaded into DataFrames for better organization. Data from both sources are merged, and irrelevant columns are dropped. This ensures that the data is in a format suitable for further processing.

Table 1
Comparison between prediction models based on multiple parameters.

Factor	Long-term dependencies	Incorporation of external factors	Non-linear relationships	Computational Complexity	Large prediction horizons
Multivariate BiLSTM	Yes	Yes	Yes	Moderate	Yes
GRU	Yes	Limited	Yes	Moderate	Yes
ARIMA	Yes	Limited	Limited	Low	Limited
SVR	Limited	No	Yes	Moderate	No
SARIMAX	Yes	Limited	Limited	Moderate	Limited
Ensemble Model (GARCH+ARIMA)	Yes	Yes	Yes	High	No
Univariate LSTM	Yes	No	Yes	Moderate	Yes

3.2.1.2. Portfolio generation. Using a Dirichlet distribution, random portfolio weights are generated for 100 portfolios initially. The Dirichlet distribution is parametrized by a vector of positive concentration parameters, and it generates samples representing the weights of each asset in a portfolio. These weights are added as columns to the DataFrame, representing each portfolio's composition.

3.2.1.3. Portfolio evaluation. Each portfolio's performance is evaluated based on predefined metrics, including expected return, downside risk, and expected ESG score. These metrics are calculated using the portfolio weights and relevant data from the DataFrame. For each portfolio, these metrics are stored in a separate DataFrame, enabling comparative analysis.

3.2.1.4. Non-dominated sorting. The non-dominated sorting function then ranks portfolios based on their performance metrics, creating a Pareto front of non-dominated solutions. This sorting algorithm ensures that no portfolio dominates another in all metrics, providing a set of diverse and optimal solutions.

3.2.1.5. Tournament selection. In tournament selection, a subset of portfolios (participants) is randomly chosen (in this case size 2), and the best portfolio among them, based on its rank, is selected as a parent for the next generation. This process is repeated until enough parents are selected, ensuring genetic diversity in the offspring population.

3.2.1.6. Crossover. Crossover involves combining the weights of two parent portfolios to generate offspring portfolios. A random crossover point is selected, and weights from each parent are swapped to create new combinations. This process introduces variability and exploration into the population. It helps to explore the solution space more comprehensively.

3.2.1.7. Multiple generations and iterations. The NSGA algorithm is run for multiple generations, where each generation involves selection, crossover, and evaluation steps. After multiple iterations, the best portfolio is identified, representing an optimal solution according to the defined criteria. This iterative process ensures continuous improvement and convergence towards an optimal solution.

3.2.1.8. Final portfolio selection. We select the best-performing portfolios based on the Maximum ESG-Sortino ratio, defined as the expected return multiplied by the expected ESG score, divided by the negative downside risk. We filter out portfolios with an ESG score below a high threshold, ensuring socially responsible investment choices. The NSGA implementation systematically and iteratively generates, evaluates, and selects portfolios guided by fitness criteria. Leveraging genetic algorithms and Pareto-based optimization, it discovers diverse and optimal portfolios tailored to specific objectives and constraints. The final portfolio is tested on 2022 stock price data and compared with the traditional Markowitz portfolio to assess the evolutionary approach's effectiveness and robustness.

3.3. Dynamic rebalancing

3.3.1. Actor critic RL model architecture

In dynamic portfolio rebalancing, Actor-Critic-RL adaptively adjusts asset allocations over time. The DRIP model, combining MBi-LSTM neural networks and MV-ESG models, has shown promise in global markets but requires adjustments for developing markets like India. This section details how Actor-Critic RL rebalances a portfolio using monthly returns and volatility data.

State Representation: The state at each time step includes current asset allocations, market conditions, historical returns, volatility, macroeconomic indicators, market sentiment, transaction costs, constraints, and investor risk preferences.

Action Space: The RL agent can buy, sell, or hold assets to rebalance the portfolio. Actions may also include alternative trading strategies or risk management techniques.

Reward Design: The reward signal guides the RL agent by providing feedback on the quality of actions. Rewards may include positive returns, risk measures (e.g., Sharpe ratio, Sortino ratio), and penalties for frequent trading or large allocation changes.

Actor-Critic Architecture: Consists of the Actor Network (Policy Network) and the Critic Network (Value Network). The actor selects actions to maximize rewards, while the critic evaluates these actions by estimating expected future rewards.

Training Phase: The RL agent interacts with the financial market over multiple episodes, exploring actions and updating policies to maximize rewards. The actor network is updated using policy gradient methods, and the critic network minimizes temporal difference error.

During rebalancing, the RL agent evaluates the current state, selects actions, adjusts portfolio allocations, and updates networks based on observed rewards. Actor-Critic RL dynamically learns optimal asset allocations in response to changing market conditions, maximizing long-term returns while managing risk.

Deploying RL-based portfolio management systems requires addressing practical challenges like data quality, computational complexity, and regulatory constraints. Further research is essential to realize RL's full potential in portfolio optimization. Fig. A3 (Appendix) represents the overall architecture for Actor-Critic-RL.

3.3.2. Reinforcement learning deployment and training

Our research implements an actor-critic reinforcement learning (RL) framework for portfolio rebalancing. The architecture includes two neural networks: the Actor Network, which uses fully connected layers with ReLU activation to determine optimal portfolio weights, and the Critic Network, which evaluates state-action pairs to guide learning. A Replay Buffer manages experiences in tuples, enabling efficient sampling for diverse data points. The Actor-Critic Agent class initializes networks, optimizers, and the buffer, and handles action selection, network updates, and transition storage.

Training involves preprocessing historical stock data to extract returns, volatility, and downside risk. The agent interacts dynamically with the environment, refining strategies based on rewards and experiences. Key hyperparameters include learning rates (0.001 for both

actor and critic networks), a discount factor (0.99), and a buffer size (10,000). These parameters influence the speed, stability, and efficiency of learning.

The agent iteratively selects actions, computes rewards, and updates networks using the replay buffer. A separate evaluation loop tests the trained agent in a simulated environment, assessing metrics such as average return, return standard deviation, maximum return, and minimum return. By adjusting hyperparameters and evaluating performance, practitioners can optimize portfolio management outcomes (see Fig. 1).

In conclusion, this sophisticated RL framework for portfolio rebalancing leverages neural networks, experience replay, and fine-tuned hyperparameters to navigate financial market complexities. The RL algorithm's performance is expected to improve over time with more training data, highlighting the potential for robust implementation and evaluation in algorithmic trading.

4. Data and analysis

4.1. Datasets and sources

Daily stock price data from Yahoo Finance for the years 2011 to 2021 (training set) and 2022 (testing set) support this study. This 11:1 split allows the model to learn from extensive historical data while testing its predictive accuracy on recent data. The use of daily stock prices provides detailed insights into market behavior, enhancing the evaluation of the model's forecasting capabilities.

4.2. Evaluation metrics

To evaluate and compare our stock prediction models, we use the MAE and RMSE to measures prediction error, preferring lower values and higher accuracy. For cross-firm comparison, MSPE and MAPE are employed to assess model efficacy in percentage terms, accounting for differences in stock price values across firms [24]. MAE quantifies the average magnitude of prediction errors by averaging the absolute difference between predicted and actual price. Lower MAE value indicates higher prediction accuracy.

Formula:

$$MAE = 1/n \sum |y_i - y'_i| \tag{1}$$

where:

- n is the number of predictions
- y_i is the actual stock price for prediction i
- y'_i is the predicted stock price for prediction i

RMSE is another measure of prediction error that penalizes larger errors more severely than smaller errors. It is calculated by taking the square root of the average squared difference between predicted and actual stock prices. A lower RMSE value indicates higher prediction accuracy.

Formula:

$$RMSE = \sqrt{\left(1/n \sum (y_i - y'_i)^2\right)} \tag{2}$$

where:

- n is the number of predictions
- y_i is the actual stock price for prediction i
- y'_i is the predicted stock price for prediction i

MAPE expresses the average absolute percentage error in relation to the actual stock price. It is calculated by taking the absolute value of the percentage difference between each predicted price and its corresponding actual price, and then averaging the absolute percentage values over all predictions. A lower MAPE value indicates higher prediction accuracy.

Formula:

$$MAPE = 1/n \sum |(y_i - y'_i)/y_i * 100| \tag{3}$$

where:

- n is the number of predictions
- y_i is the actual stock price for prediction i
- y'_i is the predicted stock price for prediction i

MSPE is like MAPE but uses squared percentage differences instead of absolute percentage differences. It is calculated by taking the average of the squared percentage differences between predicted and actual stock prices, and then taking the square root of the average [25]. A lower MSPE value indicates higher prediction accuracy.

Formula:

$$MSPE = \sqrt{\left(1/n \sum ((y_i - y'_i)/y_i)^2\right) * 100} \tag{4}$$

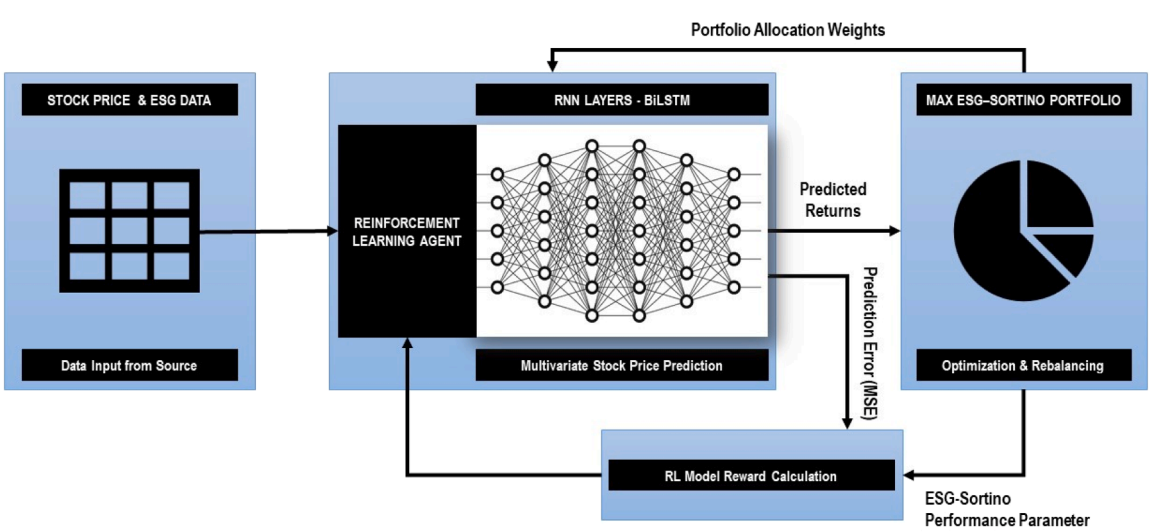

Fig. 1. Combined model architecture of the final proposed system.

where:

- n is the number of predictions
- y_i is the actual stock price for prediction i
- y'_i is the predicted stock price for prediction i

'n' in all the above formulas is 30 in this case as the number of stocks chosen are 30.

We compare the proposed MBi-LSTM's prediction accuracy against baseline and ensemble models. MBi-LSTM, which processes input sequences in both forward and backward directions, captures past and future trends, enhancing its ability to learn dependencies in both directions. This bidirectional approach is expected to improve accuracy and robustness in stock price forecasting, making MBi-LSTM superior to traditional LSTM, ARIMA, SVR, GRU, SARIMAX, and ensemble models like GARCH+ARIMA.

In portfolio allocation, we compared NSGA with conventional methods, using the MaxESG-Sortino objective function in a multi-objective optimization framework, incorporating returns, risk, and ESG score. NSGA's dynamic optimization considers returns, risk, and ESG scores, aligning portfolios with ethical values. Our evaluation compared our approach with Markowitz portfolio theory using 2022 data, analyzing returns, risk, and ESG scores. This comparison demonstrated our approach's superiority in balancing financial returns and ESG alignment.

5. Results and discussion

5.1. Stock prediction model test

The results shows that the MBi-LSTM model outperformed others and was selected as the final model based on MAE, RMSE, MAPE, and MSPE. The following Table 2 exemplifies comparative analysis of stock prediction models for HDFC Bank, encompassing MBi-LSTM, SVR, ARIMA, SARIMAX, Ensemble Model (GARCH+ARIMA), GRU and U-LSTM which reveals distinctive performance metrics. Similar comparative performance was obtained for the other four stocks tested (Table 3).

The final model design includes multiple Bi-LSTM layers with Leaky ReLU activation, dropout layers for regularization, and a final dense layer for regression, similar to [26]. Data preprocessing involved Min-Max scaling. Time series sequences for training and validation were generated using Keras' 'TimeseriesGenerator'. During training, the Adam optimizer with a learning rate of 0.001 and clipnorm set to 0.01 mitigated exploding gradients. The loss function was MSE, and early stopping was implemented to prevent overfitting as suggested by [27]. The MBi-LSTM consistently outperformed the other six models across various stocks, indicating the robustness of the multivariate approach. This enhanced predictive accuracy leads to realistic assessments of annualized returns, risks, and risk-adjusted ratios, making the MBi-LSTM model a reliable tool for optimizing investment portfolios [26]. The findings encourage practitioners and researchers to adopt the MBi-LSTM model for diverse financial forecasting scenarios.

5.2. Portfolio allocation model test

After harnessing the predictive power of MBi-LSTM models to generate anticipated returns, we used these forecasted returns to construct socially responsible investment portfolios. Using predicted returns and risk measures, we applied the NSGA technique [20] and calculated fitness value 100 randomly generated portfolios and subsequent portfolios. We set the nominal risk-free rate at 7.10 % (R_f=0.071), based on the Yield to Maturity (YTM) of 10-Year Treasury Bills issued by the Government of India [28], to integrate real-world financial consideration. Stock allocation in each portfolio was determined through the iterative NSGA optimization using the MaxESG-Sortino objective function [20]. We quantified actual annualized returns, volatility, and risk-adjusted returns using authentic stock prices from the corresponding period. Key performance metrics, including annualized returns and composite ESG Score, were systematically computed for the MaxESG-Sortino portfolio, identified as the best portfolio after the NSGA exercise. These metrics serve as quantitative benchmarks and provide qualitative insights into the portfolios' risk-return profiles and ethical considerations. The synthesis of anticipated and actual financial

Table 2
ESG scores and ratings of the 30 Indian stocks considered for our research's portfolio.

Stock Name	Industry	ESG Score	Rating
Indiabulls Real Estate Limited	Real Estate	45	Below Average
Premier Explosives Limited	Industrials	43	Below Average
Gati Limited	Logistics	43	Below Average
Apcotex Industries Limited	Chemicals - Specialty Chemicals	41	Below Average
Mahanagar Telephone Nigam Limited	Telecom	44	Below Average
Ramco Systems Limited	IT	52	Adequate
NLC India Limited	Power Thermal	52	Adequate
Sundram Fasteners Limited	Auto Ancillary	47	Adequate
Central Bank of India	Banks	58	Adequate
Dixon Technologies (India) Limited	Durables & Electricals	49	Adequate
Aarti Industries Limited	Chemicals - Specialty Chemicals	48	Adequate
Emami Paper Mills Limited	Paper	47	Adequate
Andhra Paper Limited	Paper	46	Adequate
Everest Kanto Cylinder Limited	Industrials	49	Adequate
Oriental Hotels Limited	Hotels	51	Adequate
Saregama India Limited	Media	55	Adequate
Reliance Industries Limited	Diversified	53	Adequate
Eicher Motors Limited	Auto OEM	55	Adequate
AU Small Finance Bank Limited	Banks	68	Strong
Voltas Limited	Durables & Electricals	61	Strong
JSW Energy Limited	Power Thermal	62	Strong
Mahindra Lifespace Developers Limited	Real Estate	61	Strong
Multi Commodity Exchange of India Limited	Financial Services (Others)	61	Strong
Cipla Limited	Pharmaceuticals	69	Strong
Mahindra and Mahindra Limited	Auto OEM	61	Strong
Advanced Enzyme Technologies Limited	Pharmaceuticals	61	Strong
HCL Technologies Limited	IT	71	Leadership
HDFC Bank Limited	Banks	71	Leadership
Kotak Mahindra Bank Limited	Banks	71	Leadership
Tata Consultancy Services Limited	IT	76	Leadership

Source: CRISIL 2022 ESG Ratings [9].

Table 3
Error term comparison for MBi-LSTM with other prediction models (for HDFC Bank).

	MAE	RMSE	MAPE	MSPE
MBi-LSTM	64.1046	82.4320	4.2896	0.2924
SVR	331.786	373.570	20.938	27.033
ARIMA	91.2208	111.197	6.5286	0.6471
U-LSTM	79.2982	93.5280	5.5493	0.5228
GRU	110.557	22.1236	7.4215	0.8351
GARCH+ARIMA	71.1104	93.2624	4.7584	0.3297
SARIMAX	82.3344	101.131	5.7630	0.5599

outcomes through the MaxESG-Sortino portfolios offers valuable insights. These insights inform investment decisions, refine portfolio optimization techniques, and provide nuanced perspectives for investors navigating the dynamic financial markets.

We performed mean-variance optimization for benchmarking of our NSGA produced portfolio using the predicted annualized returns for the 30 stocks portfolio and obtained the optimal weights of the Markowitz portfolio as follows (see Fig. 2).

Now with the allocations obtained through the mean-variance approach, we test the MV Portfolio on real data of 2022. We calculate the portfolio's performance measures which include real annualized returns the ESG Score. We obtained the following results Table 4.

Now, after MV Optimization, we developed the Maximum ESG Sortino Portfolio optimized and obtained after the NSGA optimization process and found the allocation weights of the 30 stocks. The obtained optimal weights are below represented Fig. 3.

With the allocations obtained through this approach, we test the proposed Portfolio on real data of 2022. We calculate the portfolio's performance measures which include real annualized returns and the ESG Score. We obtained the following results Table 5.

Now, as we have performances of both the portfolios in the year 2022, we compare both methods of portfolio optimization side-by-side in Table 6.

Form the above table, it is easily evident that our custom generated portfolio outperforms the Traditional Mean-Variance Portfolio on all parameters with a greater annualized return and finally, a greater ESG score. It is also to be noted that in 2022 the market returns were 4.3 % annualized for the Index (NIFTY50) which is lower than our portfolio and thus our model even surpassed the Index generating >1.3 % alpha.

5.3. Rebalancing model test

The RL model is still to be tested in developing economies like India to outperform non-RL techniques with the same stock prediction and portfolio optimization model as proven from a global perspective in [18]. It was expected that the model would significantly improve the model performance in terms of both the prediction of stock returns and the optimization of portfolios. However, when tested for monthly stock rebalancing for 12 months, the resulting portfolio did not yield promising results compared to our initially input NSGA portfolio similar to [29]. Following are the results and portfolio comparison for the month of January 2022, see Table 7.

Table 4
Results for Mean-Variance optimization.

MV Portfolio Real Returns 2022	−8.1971
MV Portfolio ESG Score	52.2527

However, as slight improvements were observed in the rebalanced portfolio month-on-month it is expected that when trained on a longer time horizon, the RL model portfolio can compete with the NSGA-MES portfolio in terms of obtained results (see Fig. 4). With small training data the RL model fails to identify the correct solution space and optimize the model through adaptive rebalancing, but as the model shows slight improvements which are by definition the property of a properly implemented RL mode, it is expected to perform as an efficient too in the future.

As of now, evident from the portfolio rebalancing capacity of the nascent stage RL model, self-sufficient intelligent portfolios are obviously implementable and possible, though their efficacy and accuracy are expected to improve over time with better implementation, better and complex architectures and higher quantity and quality of training data. The rebalancing model is thus a success when it comes to an idea, but obviously needs a better environment interaction and learning in context of this research's purpose.

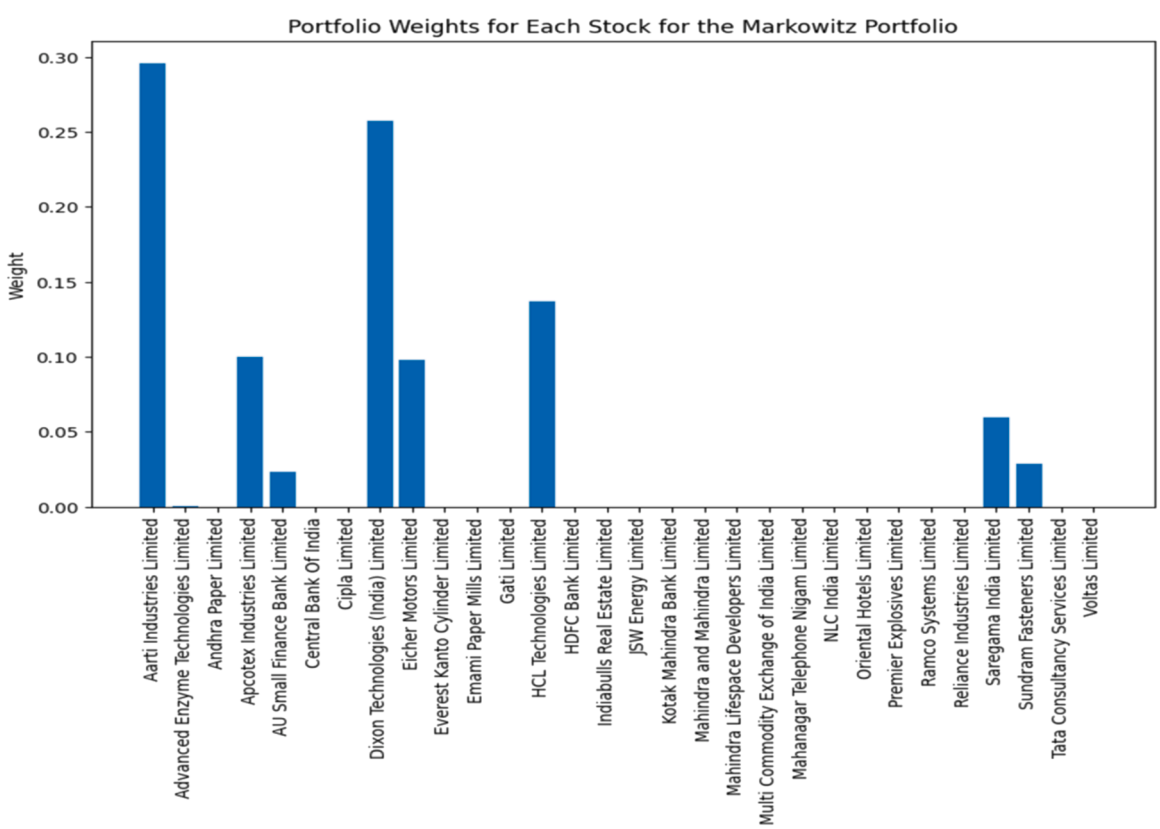

Fig. 2. Mean-variance portfolio allocation.

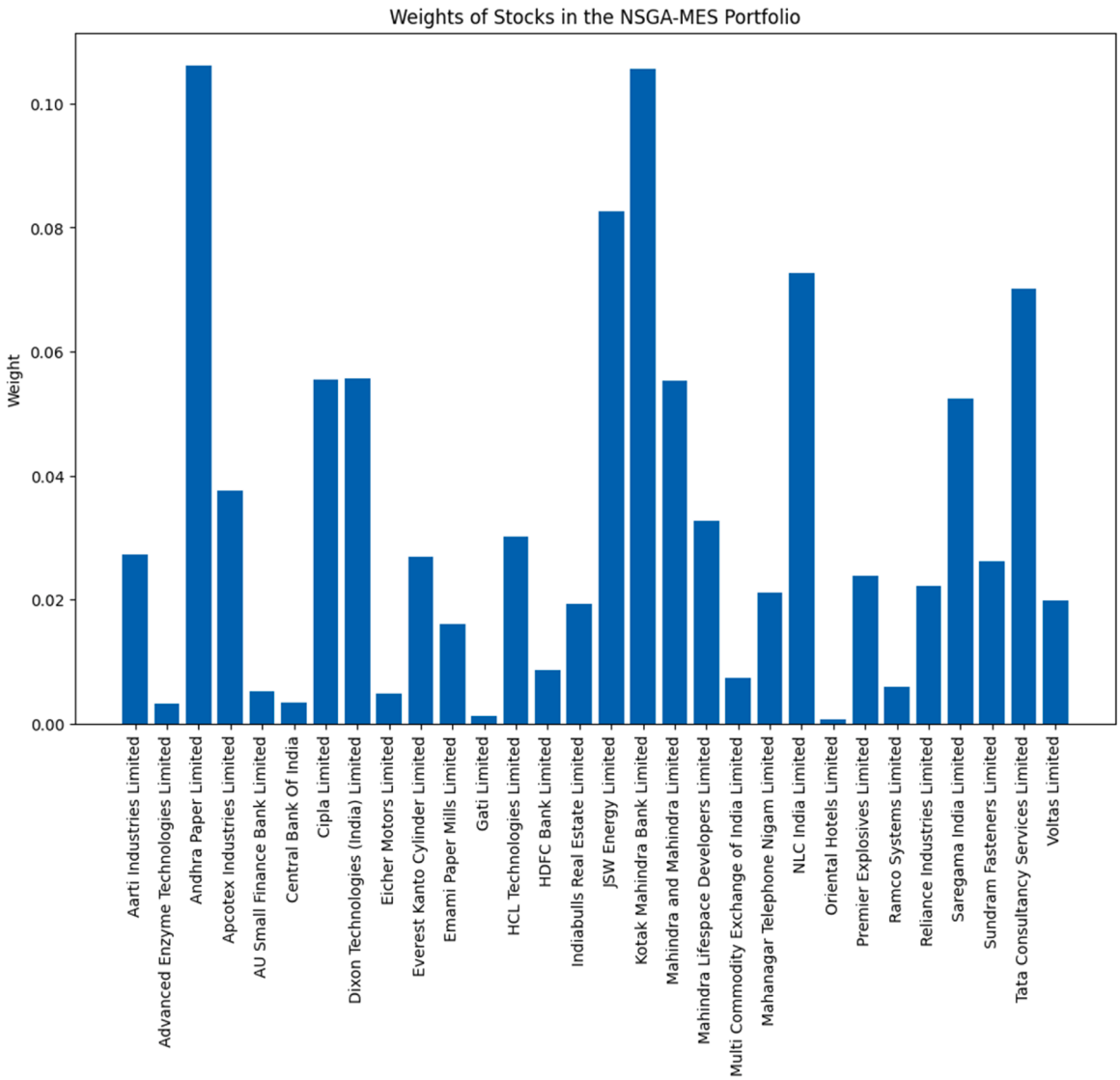

Fig. 3. Maximum ESG – Sortino ratio portfolio allocation through NSGA.

Table 5
Results for Maximum ESG – Sortino optimization.

MES Portfolio Real Returns 2022	5.6216
MES Portfolio ESG Score	57.407

Table 6
Side-by-side comparison of MV and MES portfolio optimization.

	Mean-Variance Portfolio (MV)	NSGA Portfolio (MES)	Better Performer
Portfolio Real Returns	−8.1971	5.6216	**NSGA-MES**
Portfolio ESG Score	52.2527	57.407	**NSGA-MES**

Table 7
Evaluation metrics comparison.

Metric	Agent Portfolio	Initial Portfolio
Average Return	−0.024315	0.029166
Standard Deviation	0.003912	0.000000
Max Return	−0.012412	0.029166
Min Return	−0.029002	0.029166

6. Conclusion

This study integrates modern machine learning methods with novel portfolio remodeling procedures to deeply investigate stock market dynamics. Focusing on stock prediction, portfolio optimization, and a forward-looking reinforcement learning algorithm for dynamic portfolio rebalancing, our tripartite examination offers a custom framework tailored to the Indian market's growing emphasis on sustainable and green investment.

We use advance prognostication models, primarily the Mbi-LSTM, to provide accurate financial price, return, and risk predictions. Further we employ an evolutionary optimization technique to construct a socially responsible investment portfolio and addresses the three-objective optimization problem by considering ESG. This comprehensive approach, incorporating allocation rebalancing, portfolio optimization, and stock prediction, has been validated using data from 2022, and showing superior forecasting abilities compared to standard algorithms. The MBi-LSTM model reliably predicts daily stock values of 30 prominent Indian market stocks, outperforming six other models, including sophisticated ensemble model, predictive accuracy.

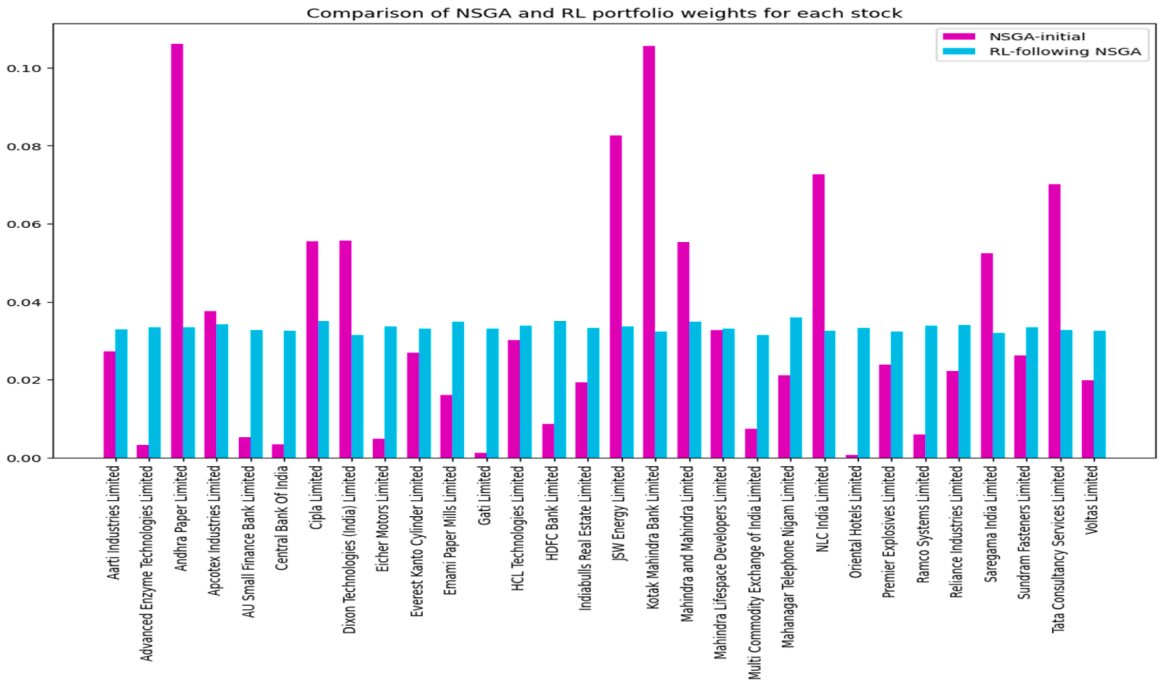

Fig. 4. Depiction of portfolio weight changes based on the RL model for Jan 2022.

Our framework, based on Bi-LSTM predictions and optimized with Non-dominated Sorting Genetic Algorithm (NSGA) using MaxESG-Sortino objective achieves high ESG ratings and competitive financial performance when compared to the standard Markowitz Mean-Variance portfolio models and the NIFTY50 index. in terms of realized annualized returns. This approach suggests a paradigm shift from traditional mean-variance optimization to iterative portfolio optimization using genetic algorithms.

With rise of socially aware investment, Indian retail investors are poised to direct capital towards firms that promote environmental sustainability and societal well-being. This paper leverages sophisticated deep-learning algorithms for stock prediction and novel reinforcement learning techniques for dynamic portfolio rebalancing, establishes a precedent for future research and the emergence of AI wealth managers. This AI managers could reduce management and transaction costs for retail investors and firms.

Our work not only adds to contemporary scholarship across disciplines, but it also has practical advantages for responsible investors, funds, as well as indexes in financial markets. We envisage greater developments in the realm of socially responsible investing as artificial intelligence matures. The research's implications are not limited to the immediate academic conversation; they also have practical applications that are in accordance with how responsible investment is developing.

7. Limitation

The presented research explores the intersection of advanced ML and portfolio optimization, suggesting future directions to enhance the proposed framework and improve the accuracy of the MBi-LSTM model's return prediction. Integrating diverse ML algorithms like GBMs or deep learning ensembles could refine risk assessment and portfolio diversification, while expanding data sources and adopting advanced prediction models could further optimize stock forecasts and rebalancing strategies. Broadening the study to include fixed-income instruments, commodities, and alternative investments would increase the model's applicability. Developing a quantitative ESG scoring framework for India, collaborating with regulatory bodies, and integrating behavioral finance insights could refine ESG assessments. Additionally, exploring the impact of external factors such as regulatory changes and geopolitical events would improve the model's adaptability. These advancements promise to reshape responsible investment practices in India by improving prediction accuracy, optimization strategies, and ESG scoring frameworks.

Ethics statements

The paper follows all the parameters provided by the journal.

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CRediT authorship contribution statement

Apurv Gaurav: Visualization, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Kripamay Baishnab:** Writing – original draft, Writing – review & editing, Formal analysis, Visualization. **Piyush Kumar Singh:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

I have nothing to declare.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.sftr.2025.100832.

Data availability

Data will be made available on request.

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